

7-1-flagship Write Equations Flagship

Equation Detective Agency | Standard: 6.EE.6 | Color-by-Number Practice

Learning Goal: I can write an equation to represent a real-world situation.

Directions: Solve each problem. Match each answer to the Color Code. On the next page, color every box with that problem number using the matching color.

Vocabulary: variable, equation, equal sign

Explain one: I solved problem ___ by ___. Because ___, my answer is ___.

Solve First

1. Write an equation: A number plus 11 is 42.

Work: _____

Answer: _____

2. Write an equation: A number minus 14 is 23.

Work: _____

Answer: _____

3. Write an equation: 7 times a number is 84.

Work: _____

Answer: _____

4. Write an equation: A number divided by 9 is 6.

Work: _____

Answer: _____

5. Write an equation: 18 more than a number is 65.

Work: _____

Answer: _____

6. Write an equation: 22 less than a number is 41.

Work: _____

Answer: _____

7. Write an equation: 12 times a number is 144.

Work: _____

Answer: _____

8. Write an equation: A number divided by 8 is 13.

Work: _____

Answer: _____

9. Write an equation: Three times a number is 75.

Work: _____

Answer: _____

10. Write an equation: A number plus 29 is 100.

Work: _____

Answer: _____

11. Write an equation: A number minus 37 is 58.

Work: _____

Answer: _____

12. Write an equation: 15 times a number is 180.

Work: _____

Answer: _____

Quick check: Before coloring, make sure every answer appears in the Color Code on the next page.

7-1-flagship Mystery Picture: Detective Badge Mosaic

Solve a problem, match the answer to a color, then color every box with that problem number.

Color Code	Color Code	Color Code	Color Code
$x + 11 = 42 = \text{red}$	$x - 14 = 23 = \text{orange}$	$7x = 84 = \text{yellow}$	$x/9 = 6 = \text{green}$
$x + 18 = 65 = \text{blue}$	$x - 22 = 41 = \text{purple}$	$12x = 144 = \text{pink}$	$x/8 = 13 = \text{brown}$
$3x = 75 = \text{gray}$	$x + 29 = 100 = \text{teal}$	$x - 37 = 58 = \text{black}$	$15x = 180 = \text{light blue}$

2	2	1	2	2	2	3	3	3	4	4	6
2	5	5	5	5	5	6	6	6	7	9	6
1	5	8	9	8	8	9	9	9	12	9	6
3	6	10	2	2	1	2	4	4	2	11	9
3	6	9	2	5	5	5	7	4	2	9	6
3	6	9	1	5	8	11	7	4	12	9	6
5	8	6	3	6	12	1	10	4	2	11	8
11	8	11	3	8	8	10	8	5	2	11	8
5	8	11	5	5	5	4	5	5	8	8	12
7	10	3	6	4	2	3	3	4	5	2	10
7	12	12	1	11	11	12	12	12	2	2	11
9	9	9	8	8	8	9	9	9	10	11	11

Early finisher writing: Choose one problem and explain your thinking in one complete sentence: Because ___, I ___.

Teacher Answer Key

#	Answer	Color	Teacher Tip
1	$x + 11 = 42$	red	Use a variable for the unknown.
2	$x - 14 = 23$	orange	Use a variable for the unknown.
3	$7x = 84$	yellow	Use a variable for the unknown.
4	$x/9 = 6$	green	Use a variable for the unknown.
5	$x + 18 = 65$	blue	Use a variable for the unknown.
6	$x - 22 = 41$	purple	Use a variable for the unknown.
7	$12x = 144$	pink	Use a variable for the unknown.
8	$x/8 = 13$	brown	Use a variable for the unknown.
9	$3x = 75$	gray	Use a variable for the unknown.
10	$x + 29 = 100$	teal	Use a variable for the unknown.
11	$x - 37 = 58$	black	Use a variable for the unknown.
12	$15x = 180$	light blue	Use a variable for the unknown.

Use note: This file is original practice aligned to the lesson skill. It is designed for low-prep review, centers, sub plans, or early finishers.